DO NOW — RECALL

Programming Techniques · Iteration and Tracing

Name: _____ Class: _____ Date: _____

1. What prints after int x=2; x=x*3;?

2. What does a queue remove first?

3. Why should loops have a stopping condition?

4. What is 0 XOR 1?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

6

2. Model answer

The oldest item

3. Model answer

To avoid non-termination

4. Model answer

1

By the end, you can…

• Distinguish count-controlled and condition-controlled loops

• Trace variables across loop iterations

• Write a terminating C# loop

Vocabulary — know these terms

Recall the term — iteration

Recall the term — accumulator

Recall the term — counter

Recall the term — condition

Recall the term — termination

NEW LEARNING

Read, annotate and explain

Trace an accumulator

int total = 0; for (int i = 1; i <= 3; i++) total += i;

Explain the important decision

Initialise the accumulator once. Each iteration changes total; i++ then advances the counter. Test i <= 3 before every pass.

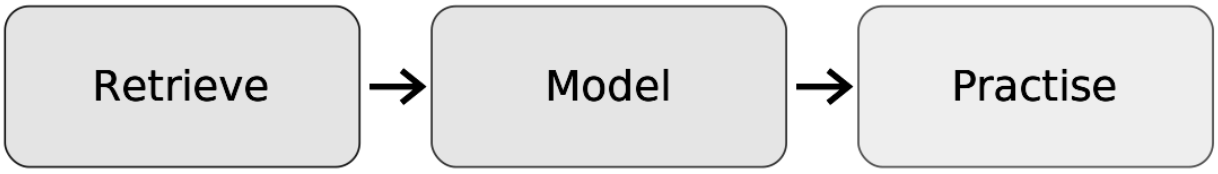
Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace an accumulator

int total = 0; for (int i = 1; i <= 3; i++) total += i;



Three-stage diagram showing retrieve, model, and practise

I DO — WORKED EXAMPLE

Follow the worked method

Trace an accumulator

Initialise and trace pass one

Initially total=0 and i=1. The test 1<=3 is true; total becomes 1, then i becomes 2.

Trace the remaining passes

At i=2, total becomes 1+2=3; i becomes 3. At i=3, total becomes 3+3=6; i becomes 4.

Show the failed test

4<=3 is false, so the body stops and total remains 6. A while loop may run zero times; a do...while tests after its body and runs at least once.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

1. Problem 1

for i = 0 to 3 exclusive, how many iterations?

Method / working:

2. Problem 2

while (x < 5) x++; starting x=3: final x?

Method / working:

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Problem 1 for i = 0 to 3 exclusive, how many iterations? Discuss, then respond

DISCUSS — Compare the sequence of steps or states. Identify the first difference in your methods and justify which step is correct.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

Guided 2: Problem 2 while (x < 5) x++; starting x=3: final x?

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Apply independently
Trace the final total for 2 + 4 + 6 using an accumulator. [3]

Question 2

Apply independently
Give a reason to choose a while loop rather than for. [2]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Apply independently
Write a C# loop to output 1, 2, 3. [3]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

True or false: a while loop always runs at least once.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Check your thinking

3: i=0,1,2.

IMPROVE: Explain one correction to your own reasoning.

Guided 2 — original CHECK

Discuss, then respond

while (x < 5) x++; starting x=3: final x?

Check your thinking

5.

IMPROVE: Explain one correction to your own reasoning.